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Smart traffic management system using IOT and machine learning approach

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***Abstract*— As we know that due to heavy population in urban areas, our cities are dealing with many problems like pollution, water shortages, traffic jams etc. So, overcome this Situation there is a concept comes in role that is "Smart City". The main purpose of Smart City is to create a society which can perform effectively and efficiently making effective use of city infrastructures through machine learning and artificial intelligence. It also focuses to optimize city functions and drive economic growth while improving quality of life for its citizens using smart technology. Smart City makes use of Artificial Intelligence, machine learning and Internet of Things (IOT) devices such as connected sensors, lights, and meters to collect and analyze data. The cities then use this data to improve infrastructure, public utilities, services and humans are interact with different devices like Smart homes , smart health , smart vehicles , smart agriculture etc.**

Machine learning will help the power for control the autonomous vehicles or self-driving vehicles to reduce delays in traffic and to reduce pollution emission by using e-vehicle.

IOT based Intelligent Transportation Systems make the exchange of information possible through cooperative systems that broadcast traffic data to enhance road safety. Traffic light assistance systems in particular utilize real-time traffic light timing data by accessing the information directly from the traffic management center. To test the reliability of a traffic light assistant system based on networked inter vehicular interaction with infrastructure, we present in this paper an approach to perform theoretical studies in a lab-controlled scenario. The proposed system retrieves the traffic light timing program within a range in order to calculate the optimal speed while approaching an intersection and shows a recommended velocity based on the vehicle's current acceleration and speed, phase state of the traffic light, and remaining phase duration. Results show an increase in driving efficiency in the form of improvement of traffic flow, reduced gas emissions, and waiting time at traffic lights after the drivers adjusted their velocity to the speed calculated by the system.

Keywords –Machine learning , IOT, smart vehicles, Intelligent Transportation.

I. INTRODUCTION

By 2050 70% of the world population will live in 2%ofthe Earth's surface. It is very challenged for us. Trends continue to grow; this raises many issues like pollution, infrastructure, access, traffic congestion, mobility, safety and health. Development and combination of new technology such as IOT and AI offers a solution is Smart City.

'Smart' is related to actually develop smart application that make day to day living for us comfortable. Smart city is related to promote cities that provide core infrastructure and give a decent quality of life to its citizens, a clean and sustainable environment of "smart" solution. We want to provide our know-How to enable public authority, business and society as a whole to better anticipate the future in creating these cities tomorrow. Machine learning is used to grasp data/exploding data and gives local player, citizen, entrepreneurs to growth of smart city.

Example: in United Kingdom city BRISTOL created data dome. It is a public platform where information of traffic, air quality, noise, governance can be shared.

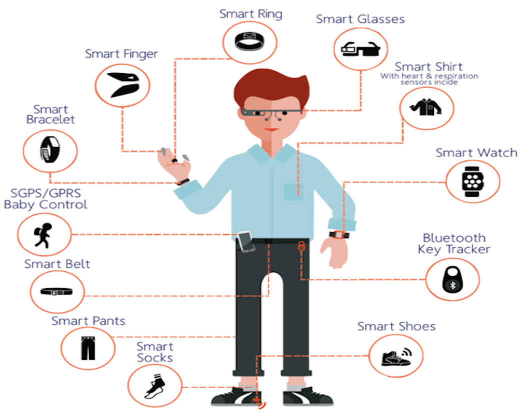
A. SMART CITY: Smart city is a city that connects citizens using digital technology. And make their living betters. IOT source censors, video, camera, social media and some others inputs like a nervous system. Those who give constant feedback to administrators and citizens, so that they can take better decisions. Smart city is one of the powerful applications of source. It is estimated that by 2050, 65% of the world population will live in towns. The town infrastructure is not growing according to the population in the town. Infrastructure like Transportation, electricity etc. Its demand is growing and it will get worse with time. Smart city also solves traffic problems, Smart city also makes public transportation better, Smart city also improves waste management, Smart cities also improve public safety.

***B. IOT*:** The internet of things, or IOTa, is a system of interrelated computing devices, mechanical and digital machines, objects, animals or people that are provided with unique identifiers (UIDs) and the ability to transfer data over a network without requiring human-to-human or human-to-computer interaction. A thing in the internet of things can be a person with a heart monitor implant, a farm animal with a biochip transponder, an automobile that has built-in sensors to alert the driver when tire pressure is low or any other natural or man-made object that can be assigned an IP address and is able to transfer data over a network. Increasingly, organizations in a variety of industries are using IOT to operate more efficiently, better understand customers to deliver enhanced customer service, improve decision-making and increase the value of the business.IOTa evolved from machine-to-machine (M2M) communication, i.e., machines connecting to each other via a network without human interaction. Machine-to-machine refers to connecting a device to the cloud, managing it and collecting data. Taking machine-to-machine to the next level, IOT is a sensor network of billions of smart devices that connect people, systems and other applications to collect and share data. As its foundation, M2M offers the connectivity that enables IOT.

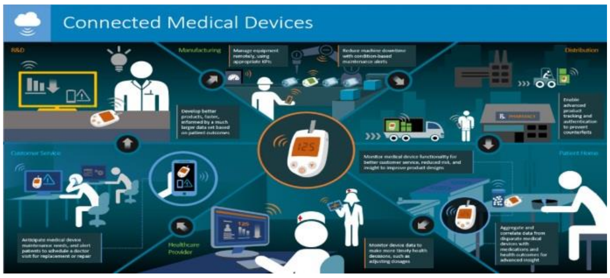
APPLICATION: There are numerous real-world applications of the internet of things, ranging from consumer IOT and enterprise IOT to manufacturing and industrial IOT (IIOT). IOT applications span numerous verticals, including automotive, telecom and energy.

In the consumer segment, for example, smart homes that are equipped with smart thermostats, smart appliances and connected heating, lighting and electronic devices can be controlled remotely via computers and Smartphone.

Wearable devices with sensors and software can collect and analyze user data, sending messages to other technologies about the users with the aim of making users' lives easier and more comfortable. Wearable devices are also used for public safety -- for example, improving first responders' response times during emergencies by providing optimized routes to a location or by tracking construction workers' or firefighters' vital signs at life-threatening sites.



In healthcare, IOT offers many benefits, including the ability to monitor patients more closely to use the data that's generated and analyze it. Hospitals often use IOT systems to complete tasks such as inventory management, for both pharmaceuticals and medical instruments

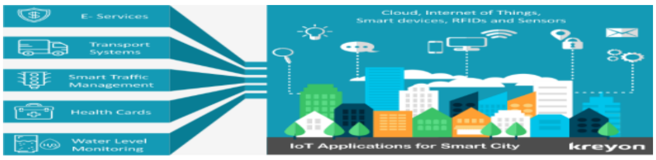


In Smart buildings, for instance, reduce energy costs using sensors that detect how many occupants are in a room. The temperature can adjust automatically -- for example, turning the air conditioner on if sensors detect a conference room is

full or turning the heat down if everyone in the office has gone home.

In agriculture, IOT-based smart farming systems can help monitor, for instance, light, temperature, humidity and soil moisture of crop fields using connected sensors. IOT is also instrumental in automating irrigation systems.

In a smart city, IOT sensors and deployments, such as smart streetlights and smart meters, can help alleviate traffic, conserve energy, monitor and address environmental concerns and improve sanitation.



C. MACHINE LEARNING: It is a type of Artificial Intelligence that provides computer with the ability to learn without being explicitly programmed. Machine Learning focuses on the development of computer program that can change when exposed to new data.

TYPES OF LEARNING:

(i) Supervised Learning – It is a type of machine learning algorithm that uses a known dataset (called the training dataset) to make predictions, The training dataset includes input data and response values. From, it the supervised learning algorithm seeks to build a model that can make predictions of the response values for a new dataset.

(ii) Unsupervised Learning - It is a type of machine learning algorithm used to draw inferences from datasets of input data without labeled responses.

(iii) Reinforcement Learning - It is an area of machine learning inspired by behaviorist psychology, connected with how software agents ought to take actions in an environment so as to maximize some notion of cumulative reward.

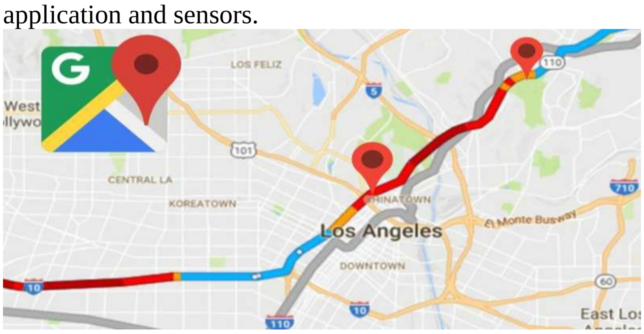
ALGORITHMS: Classification Algorithm, Anomaly Detection Algorithm, Regression Algorithm, Clustering Algorithm, Reinforcement Learning.

II. RELATED WORKS

Google map: Machine learning is used in Google map for faster route selection. It is used for traffic prediction. Map consist combination of different color signifies different traffic level of clear, slow moving or heavily congested. Machine learning is the key to providing location data in a scalable way. It is to enable the automatic extraction of information from geo-located imagery to improve Google Maps. It uses neural network that automates the image information reading process. This algorithm is publicly available on GitHub through TensorFlow, which is Google's own open-source machine learning software library. Google is already implemented machine learning to identify car license plates and many other. And now, it is using the same technology to fetch information from street, road, traffic. Using this technology, Google aims to improve the location data of the world's addresses. The latest Machine Learning algorithms helped achieve 84.2 % accuracy when tested on several challenging street signs in France. These statistics significantly outperformed the previous state-of-the-art systems.

We can find it by two measures.

1. 'Average time taken' on the specific days at specific times on that route.
2. 'Real time location' Data of vehicles from Google maps application and sensors.



Traffic signal: A traffic signal installation comprises a controller, traffic light heads and detection. The controller is the 'brain' of the installation and contains the information required to force the lights through various sequences. Traffic signals can run under a variety of different modes which can be dependent on location and time of day.

Camera: Lots of traffic cameras installed on the roads which gives pictures of roads and there speed cameras which senses the speeds. Traffic light cameras at busy junction play a vital role in keeping our roads safe. But if you've accidentally run a red light, it may not be clear whether or not you've been caught. Traffic light cameras detect vehicles which pass through lights they've turned red by using sensors or ground loops in the road. speed camera, fixed speed camera, mobile speed cameras.

Conducting loop detector: Apart from this loop –detector parts which hardly follow the road like a strap or something buried under the road when any vehicle crosses or passes through the loop detector it count the number of vehicles crossing through the loop detector and it gives us an approximate measure of how many vehicle has crossed this loop detector. Its installation is quick and easy. Its adjustment ranges of sensitivity are from 0 to 9. Sensitivity simply adjusts the sensitivity to the value shown on the display when a vehicle is position on the loop. Electrically conducting loop which is insulated is installed in the pavement.

III. PROPOSED WORK

Proposed system shown in the below figure 1 is design for establish traffic at road lines, sensing through sensors, cameras speed cameras and RFIDs which are work well defined manner .It processes sensors' data at the node level and videos and data are store in the local server and also calculate the cumulative density of the vehicles. This things are also handle the emergency vehicle like ambulance, fire brigade and also help to known how much traffic congestion are at the road through the local data. There are three types of proposed system. (i) Data Acquisition and collection layer (ii) Data Processing and Decision –making layer (iii) Application layer Actuation layer.

A. Data Acquisition and Collection Layer

Different traffic detection techniques were used by the Researchers in the state of the art which mainly consider resources like ultrasonic sensors, RFIDs, surveillance cameras and light beam. All these have own merits and demerits. Above of these surveillance cameras, ultrasonic sensors, RFIDs, smoke sensors and flame sensors are mostly used for proposed system. A surveillance camera is mostly used to detect the road traffic due to its efficiency and ease of maintenance. Blob detection algorithm has capability of noise reduction due to which it is used in video stream but at the local server. After detecting traffic, density is send to the respective microcontroller by the local server which is measured through image processing. Most of the traffic management system applications use sensors to detect density due to its enhancement the accuracy. It is used to measure distance by sending sound wave of a specific frequency and listening to bounce back. It measures the distance 2 cm to 400 cm.

Distance is calculated by using the following formula:
Distance = ((a x b) / 2) …………….(1)
a= Speed of sound, b= time taken

In this technique, there is use of three pairs of sensors which is embedded on every roadside of an intersection at certain distance to calculate the traffic density. Each sensor reads either 1 or 0. Density is calculated at the node side.

$$\sum_{i=1}^{3}(Pi) = P_i + P_{i+1} + P_{i+2} \quad ....(2)$$

P is the pair of ultrasonic sensors. Table 1, shows the states of the sensors and their results are as follow:

TABLE 1: *traffic density states by ultrasonic sensors*

Condition/Sensors	***P1***	***P2***	***P3***	***Status***
Condition 1	*1*	*0*	*0*	*Low*
Condition 2	*1*	*1*	*0*	*Medium*
Condition 3	*1*	*1*	*1*	*High*

Microcontroller receives all results from sensor and video from a local server to calculate cumulative density using Table 2.

TABLE 2: *cumulative density*

Situations	**Sensors' Result**	**Camera's Result**	**Traffic Density**
Situation 1	High	High	High
Situation 2	High	Medium	High
Situation 3	High	Low	Medium
Situation 4	Medium	High	High
Situation 5	Medium	Medium	Medium
Situation 6	Medium	Low	Medium
Situation 7	Low	High	Medium
Situation 8	Low	Medium	Medium
Situation 9	Low	Low	Low

B. Data Processing and Decision-making layer

Data Processing and Decision-making layer is the second layer of the proposed system .This layer are manage the traffic rule according to the condition of traffic signal .In the first layer of the proposed system each traffic signal has predetermined time i.e. α second, when road has normal traffic . Every signal is departure to green at their twist for α second and other signal at that time signal is red until another traffic signal is completed their turn. In the present scenario fixed-time signal control system is not work properly because day by day traffic ratio signal are increased In this time there is horrific need to count a density based traffic management model which added the time dynamically to each road according to the traffic density .In the below algorithm second part of algorithm when the capacity of traffic is not in routine .The System calculate the level of density according to the above Table 2 and update the β time of traffic signal according to the traffic density.

TABLE 3 : Algorithm

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Part(I) When no emergency vehicle detected
if (Traffic Density == high)
   if (Rush Interval == Yes)
      Time = ( ( α e^x sinθ ) + β)+γ
   else
      Time = ( α e^x  sinθ ) + ( cosθ * γ ) + β
Else
   if (Rush Interval == Yes)
      Time=(α e^x  sinθ ) +  γ
   else
      Time = α

Part (II) RFID tags detect emergency vehicle

   While (vehicle Exits)
      Time ! = 0
```

Where α is regular prefixed time given to a specific roadside, θ=90, x=0, β is extra time added in case of traffic congestion and Y is extra time added when there is rush interval near to approach.

C. APPLICATION AND ACTUCATION LAYER

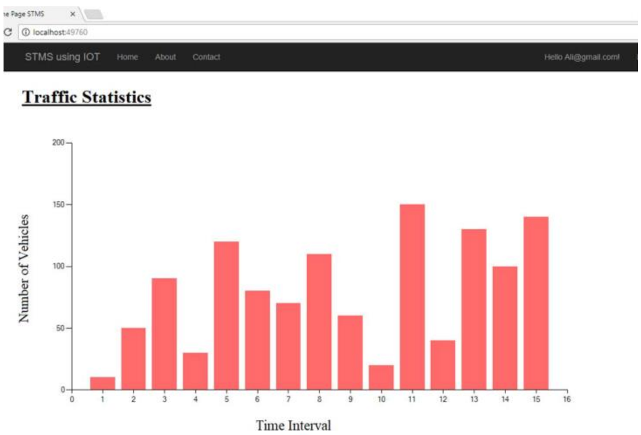
There are two types of information delivered of Actuation layer which is first duration of a green signal from system to traffic signal. And second is daily, weekly, monthly and yearly reports to the Administration of traffic management system by the web application from a centralized server. The system calculates rush interval by using regression tree algorithm on the information saved on the web application server. And this information is updates through the centralized server on the 24 hours on the daily days.

The rush interval is time span of the thirty minutes. The rush interval is then displayed on the basis of web application which is linked to a centralized server, centralized server which is the administration of smart traffic management system that displayed at daily, weekly, monthly information is successful for the road planning and resource management system. In Actuation layer, the rush interval is identified, the local server intimates to the respective microcontroller along with the road id. After the receiving the information the decision making layer reports the duration of the green signal to the respective traffic signal. In modern area, where time is money and wastage of time are not modest ,there is a need to know the people traffic condition on the road by using mobile application and digital technology which is help to solve the problem. This system is also capable of managing emergency situations like if the smoke and fire detect on the road.

In case of the road , which is detect the flame sensors and extensive smoke by smoke sensors, the system intimates to the relevant department through a mobile application for the future purpose. A traffic signal is a signaling device located at road intersection for regulating traffic from a different direction to avoid accidents and to maintain a smooth flow of vehicles. Traffic sign are very important for examination purpose and also for ready safety in India. Traffic signal control like traditional solutions that is more traffic signal or option pre-timed control. Alternatives solution of traffic signal that is Improve public transportation systems, implement transportation policies and restrictions, Autonomous car , traffic-responsive control of traffic signals. Dynamic traffic signal control i.e. dynamic signals are programmed to adjust their timing and phasing to meet changing traffic conditions. Bar graph is representing real-time traffic data. Different bar graphs are representing historical and real time data are being drawn in this traffic application which is helpful for departments and other related authorities which is progress in future planning and managing traffic congestions.

D. Result and Discussion

In this article there is a system proposed for road safety, In other words we can say that it is a prototype to demonstrate that how we can solve our road traffic issue and make our road safety. It states that at the certain points there is a camera which monitors the traffic density and calculate the vehicles in the road, and maintain some distance between them. If the traffic density crossed through the specified threshold on a road, the system stopped the normal operation and kept the green light on till the situation on the road became normal. The system monitors that, how many vehicles in the road and maintains some distance and sent real time information to the central server. Besides this, a web interface was also developed for the authorities to show them the statistics of traffic on the roads so that they could make real-time and future decisions as discussed in the statistical traffic data i.e. number of vehicles passed in a particular time span at a particular road, or we can say that system monitors when traffic is so high and when traffic is less to those stats is shown below.



IV. Conclusion and Future Work

With the advancement of emerging technology, industrial and educational development there are more opportunity of employment and better scope of education as well as research in developing cities. The life style of people in metro cities with large volume of population is equally affected by various application and service systems. Therefore currently most of the cities are in the process of transforming into smart cities by adopting automated systems in all possible sectors. With an objective of developing a new transport system for vehicles in a smart city, this article proposed a modern traffic control system using connected vehicle technology under different configuration with an integrated approach of solving general traffic related issues in a high volume traffic gateway. For the overall benefit of the traffic system, various modules like video monitoring, smart traffic control system, signal system and smart devices are included in the presented approach with detailed structure of their smart functionality. Simulation results show that it has an improved rate of congestion control in traffic points as it uses advanced technology of automating vehicles, mobile agent and big data analytic tools.

Vehicle to vehicle communication is currently not available, but it will happen in future when we speak about autonomous transportations vehicles will drive themselves and they communicate with each other and also with the infrastructure to make decision on this speed they are applying break or when to change overtaking one vehicle all these stuffs. The future scope for the project is identifying the road accident using camera, sensor and video processing. We can have a software system which can identify the overlapping image in real time video streaming and condition of accident can be decided with overlap of image. Overlap occurs then we can use SMTP to send mail directly to the government hospitals.

The project is implemented in the following way, First the ultrasonic sensors are fixed on each lane. They have used three types of sensors: High, Medium, and Low. High is sending on 1st priority Medium on 2nd and Low on 3rd priority respectively. Data is collected from sensors and sent to the system. Traffic density is measured and average waiting time is calculated

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